

PROMPT FOR DETERMINING THE EXACT RAMSEY NUMBER $R(5, 5)$

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ABSTRACT. This document contains a full frontier-model research prompt for determining the exact diagonal Ramsey number $R(5, 5)$. The current rigorous range is $43 \leq R(5, 5) \leq 46$; the task is to collapse this range to one integer with independently checkable certificates. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

For integers $s, t \geq 2$, the Ramsey number $R(s, t)$ is the least n such that every simple graph on n vertices contains either a clique of size s or an independent set of size t . Equivalently, a graph is a $(5, 5)$ -Ramsey graph if neither it nor its complement contains K_5 .

Resolve the exact value of $R(5, 5)$ completely. As of the status date for this prompt, the rigorous bounds are

$$43 \leq R(5, 5) \leq 46.$$

Thus the only possible values are 43, 44, 45, and 46. Assume for purposes of this task that a complete resolution is reachable. Do not assume which value is correct.

Complete-resolution standard

A complete solution must prove $R(5, 5) = N$ for one $N \in \{43, 44, 45, 46\}$ by establishing both sides of the equality:

- a lower bound $R(5, 5) \geq N$, normally by giving a simple graph on $N - 1$ vertices with no K_5 and no independent 5-set, or by an equally rigorous theorem implying such a graph exists;
- an upper bound $R(5, 5) \leq N$, by proving that no $(5, 5)$ -Ramsey graph on N vertices exists;
- for every computational component, a reproducible exact certificate and an independent verifier whose soundness is explained mathematically;
- a proof that every symmetry reduction, canonical augmentation rule, pruning inequality, degree restriction, and case split is exhaustive and cannot discard a valid Ramsey graph.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- improving only one side of the interval while leaving two or more possible values;
- giving a graph whose clique and independence numbers were checked only heuristically or by floating-point software;
- a SAT or constraint-programming claim of unsatisfiability without a checkable proof certificate, or with symmetry-breaking clauses whose equisatisfiability is not proved;
- enumeration through a fixed subclass unless every possible $(5, 5)$ -Ramsey graph at the target order is rigorously shown to lie in that subclass;
- a probabilistic construction that does not yield the required exact lower-bound graph or theorem;
- a reduction to an unproved classification or an unavailable proprietary computation.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- local structure around vertices and edges, using exact catalogues and constraints for $R(4,5)$ and related smaller Ramsey classes;
- linear-programming inequalities for degree sequences, common-neighbor counts, subgraph counts, and graph-complement duality;
- canonical generation, isomorph rejection, graph gluing, neighborhood extension, and independently implemented exhaustive enumeration;
- SAT, SMT, pseudo-Boolean, constraint-programming, and integer-programming encodings with DRAT, LRAT, VeriPB, or comparably checkable certificates;
- flag-algebra or semidefinite inequalities converted to exact rational certificates and combined with finite structural case splits;
- construction search for orders 43-45 using circulant, Cayley, block, switching, algebraic, coding-theoretic, local-search, and evolutionary ansatzes;
- new structural theorems that force a bounded collection of neighborhoods, separators, equitable partitions, or degree patterns at the remaining orders;
- formalization of the non-computational skeleton in Lean, Isabelle, or another proof assistant, with small trusted checkers for generated certificates.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- confirm the graph is simple, has exactly the claimed number of vertices, and is parsed identically by all verifiers;
- check every 5-vertex subset for being a clique in the graph or in its complement, using at least two

independent implementations for a lower-bound witness;

- audit off-by-one logic: a witness on $N - 1$ vertices proves only $R(5, 5) \geq N$, while nonexistence on N vertices proves only $R(5, 5) \leq N$;
- prove that complement symmetry, degree bounds, pointed-neighborhood catalogues, and isomorphism pruning preserve all cases;
- validate every unsatisfiability certificate with a small independent checker, and test the encoding on known satisfiable and unsatisfiable smaller instances;
- reproduce all catalogue hashes, graph counts, intermediate case counts, random seeds, compiler versions, and source code needed for replication;
- conduct a novelty audit so that the final claimed exact value and essential argument are not merely rediscoveries of an existing unpublished or obscure computation.

Required final artifact

The final response must contain a conventional mathematical proof and a compact archival package. At minimum include:

- the exact integer N and a logically complete proof of $R(5, 5) = N$;
- any lower-bound graph as an adjacency matrix, graph6/sparse6 string, and edge list, with cryptographic hashes;
- any upper-bound certificate in a standard independently checkable format, plus the generator and checker source;
- a manifest describing every file, the exact verification commands, expected outputs, and resource requirements;
- an independent verification report produced by an agent or implementation that did not generate the candidate.

Public search may be used for ordinary mathematical background, standard named theorems, and the published 43 and 46 bounds. Do not search the public web for a purported new exact value or answer this task merely by reporting the current interval.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.